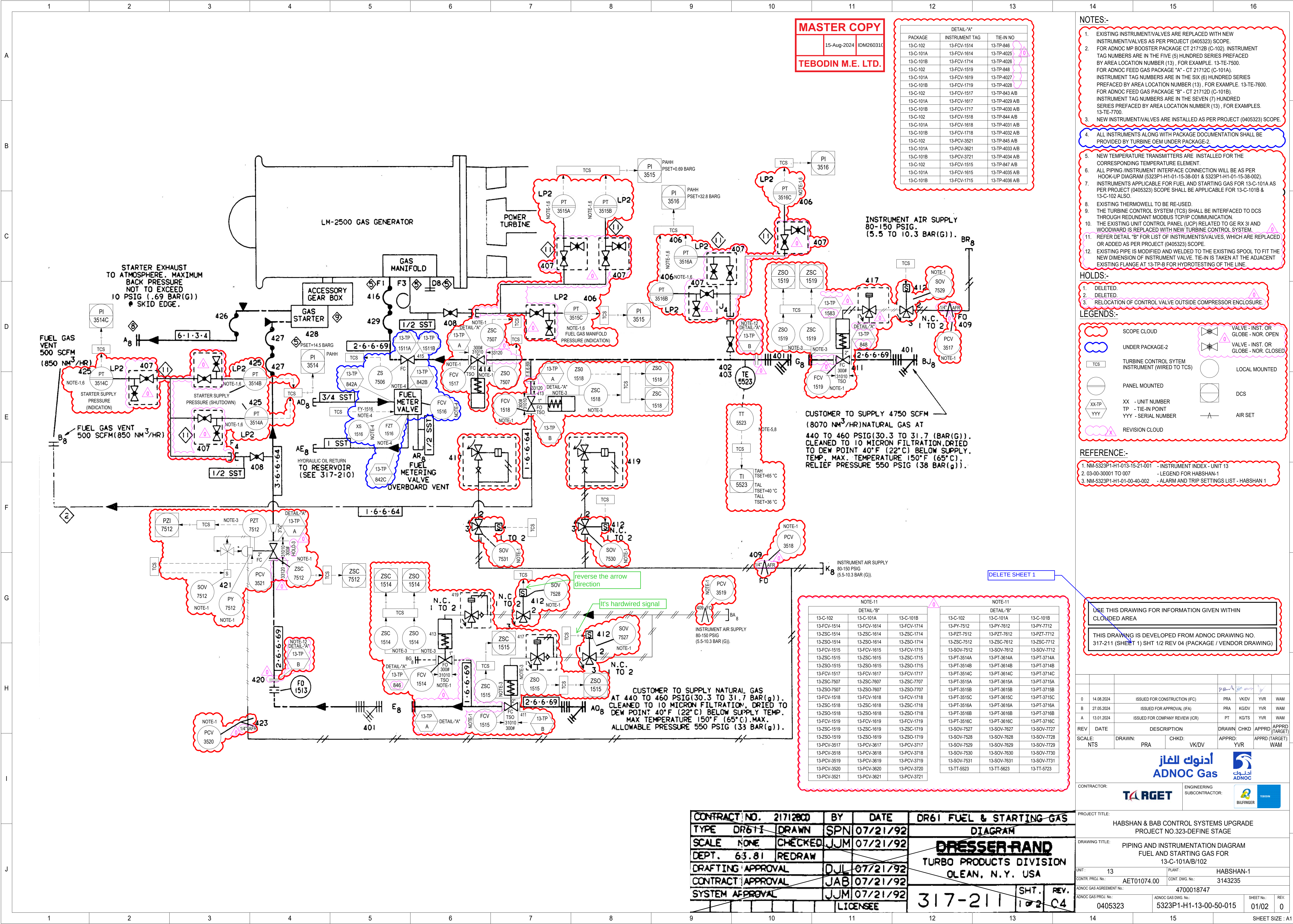


			ADNOC GAS HABSHAN & BAB CONTROL SYSTEMS UPGRADE (PACKAGE 1) PROJECT NO. 0405323						  				
			COMMENTS RESOLUTION SHEET										
Document Title:			PIPING AND INSTRUMENTATION DIAGRAM FUEL AND STARTING GAS FOR 13-C-101A/B/102				AGP Response Code			Prepared By:	VK		
Document No:			5323P1-H1-013-00-50-015		Revision No:	B	1. Revise and Re-submit		3. Approved	✓	Checked By:	DV	
AG Ref. No:			ADNOCGP-WTRAN-007148		Date:	3-Jun-2024	2. Approved with comments		4. For Information only		Approved By:	YVR	
SN	Reference		COMPANY Comments				CONSULTANT Response			COMPANY Resolution			
	Page No.	Clause No.											
1	-	-	No Comments.				-						
2	sht 1 of 2	-	-				Following changes are made in P&ID w.r.t. Rev.B submission as per piping/instrumentation inputs: 1. Instrument manifold valves for all PT are shown in project scope cloud based on instrument input. 2. On-Off valve FCV-1514 Tie-in location updated based on piping input. Hence, Tie-in number TP-8846A/B is updated as 13-TP-846. 3. On-Off valve FCV-1515 Tie-in location updated based on piping input 4. On-Off valve FCV-1519 Tie-in number updated based on piping input. Hence, Tie-in number 13-TP-A/B is updated as 13-TP-848 & 13-TP-1583. 5. Control valve PCV-3521 inlet Tie-in location updated based on piping input. piping class is also updated based on piping input. 6. Control valve 13-PCV-3517/ 3518 and PCV-3520 size is updated based on piping input. 7. Control valve FCV-1517 and FCV-1518 inlet Tie-in location updated based on piping input. piping class is also updated based on piping input. 8. Note 11 and 12 added. 9. Hold 3 added. 10. Table to mention instruments in project scope for 13-C-102, 13-C-101A and 13-C-101B is added. Refer detail "B".						

NOTICE: THIS DRAWING AND THE INFORMATION IT CONTAINS IS THE SOLE PROPERTY OF ADNOC GAS PROCESSING. IT SHALL NOT BE REPRODUCED OR LENT WITHOUT THE AGREEMENT OF THE OWNER



SCAVENGE OIL TO GAS
GENERATOR OIL CONSOLE
(SEE DIAGRAMS 317-209
AND 317-210)

SCAVENGING AIR FROM GAS GENERATOR ON-ENGINE EDUCTOR SYSTEM. CONNECTION TO BE MADE AFTER STARTER IS INSTALLED ON ACCESSORY GEARBOX.

STARTER OIL DRAIN TO CONNECTION ON ACCESSORY GEARBOX. CONNECTION TO BE MADE AFTER STARTER IS INSTALLED ON GEARBOX.

STARTER LUBE OIL SUPPLY
FROM GAS GENERATOR ON-ENGINE
LUBE OIL SYSTEM. CONNECTION
TO BE MADE AFTER STARTER
IS INSTALLED ON ACCESSORY
GEARBOX.

NOTE-15
SE
7515 SPEED SENSOR
GG STARTER
SPEED PICKUP

1. FOR COMPONENT DESCRIPTION SEE BILLS OF MATERIALS # 317-211-01, 317-211-02, 317-211-203, 317-211-204 AND 317-211-203.
2. VENTS TO HAVE MAXIMUM BACK PRESSURE NOT TO EXCEED 2 PSIG (14 BAR G).
3. SYMBOLS & LETTER DESIGNATIONS PER DRESSER-RAND DRAWING 478-373-001.
4. PIPE LINE CODES PER DRESSER-RAND DRAWING 003-152-001.
5. SEE GENERAL ELECTRIC DRAWING L31705 FOR DESCRIPTION OF CONNECTIONS ON GAS GENERATOR.
6. FOR CUSTOMER CONNECTION LETTER DESCRIPTION SEE DRAWING 478-139-601.
7. MECHANICAL STOP ON STARTER CONTROL VALVE MUST BE ADJUSTED TO LIMIT MAXIMUM CRANK PRESSURE TO 210 PSIG. ADJUSTMENT SHOULD BE MADE WITH IGNITORS AND FUEL SUPPLY DISABLED.
8. STARTER EXHAUST CONFIGURATION SHALL PROVIDE PROTECTION FROM THE ENVIRONMENT.
9. STARTER PERFORMANCE.

FUNCTION	SPEED	GAS FLOW
PURGE	1200 RPM	640 SCFM (1088 NMHR)
WATER WASH	1200 RPM	640 SCFM (1088 NMHR)
ESD MOTORING	2050 RPM	2200 SCFM (3740 NMHR)
CUT OUT	4500 RPM	1830 SCFM (3111 NMHR)
10. FILL STARTER SUMP WITH SYNTHETIC OIL PRIOR TO INITIAL START.
 11. EXCEPT DURING TESTING, INSTRUMENT BLOCK VALVES IN SENSING LINES TO TRANSMITTERS PT-3514 AB, PT-3515 AB AND PT-3516 AB MUST BE LEFT OPEN WHILE UNIT IS OPERATING. OPERATION WITH A CLOSED BLOCK VALVE COULD DISABLE THE SHUTDOWN FUNCTION WITH POSSIBLE RESULT ANT DAMAGE TO THE GAS GENERATOR AND INJURY TO PERSONAL.
12. REFERENCE DRAWINGS,

GAS TURBINE WIRING DIAGRAM	# 478-130
UTILITY LIST	#478-554
INSTRUMENT INDEX - CT 21712B	#003-175-151
CT 21712C/D	#003-175-152
LOCAL PANEL GENERAL ARRG1-LP1	# 478-450-001
LOCAL PANEL WIRING DIAGRAM-LP1	#478-131
TURBINE SKID PS & GP GENERAL ARRANGEMENT LP2	#478-127
TURBINE SKID PS & GP WIRING DIAGRAM-LP2	#478-132
OUTLINE AND FOUNDATION DRAWING, CT 21712B	#478-144
CT 21712C/D	#478-241
13. PIPING PER D-R SPEC. 003-230-001 AND DEVADED 003-230-017
14. FOR ADNOC MP BOOSTER PACKAGE CT 21712B INSTRUMENT TAG NUMBERS WILL BE IN THE FIVE (5) HUNDRED SERIES PREFACED BY AREA LOCATION NUMBER (13). E.G 13-17E-7500.
FOR ADNOC FEEB GAS PACKAGE 'A' CT 21712C. INSTRUMENT TAG NUMBERS WILL BE IN THE SIX (6) HUNDRED SERIES PREFACED BY AREA LOCATION NUMBER (13). E.G 13-17E-7000.
FOR ADNOC FEEB GAS PACKAGE 'B' CT 21712D. INSTRUMENT TAG NUMBERS WILL BE IN THE SEVEN (7) HUNDRED SERIES PREFACED BY AREA LOCATION NUMBER (13). E.G 13-17E-7700.
15. ALL INSTRUMENTS ALONG WITH PACKAGE DOCUMENTATION SHALL BE PROVIDED BY TURBINE OEM UNDER PACKAGE-2.
16. THE TURBINE CONTROL SYSTEM (TCS) SHALL BE INTERFACED TO DCS THROUGH REDUNDANT MODBUS TCP/IP COMMUNICATION.
17. THE EXISTING UNIT CONTROL PANEL (UCP) RELATED TO GE RX 31 AND WOODWARD IS REPLACED WITH NEW TURBINE CONTROL SYSTEM.

	UNDER PACKAGE-2
	REVISION CLOUD
	SCOPE CLOUD

USE THIS DRAWING FOR INFORMATION GIVEN WITHIN CLOUDED AREA

THIS DRAWING IS DEVELOPED FROM ADNOC DRAWING NO.
317-211 (SHEET 2) SHT 2/2 REV 04 (PACKAGE / VENDOR DRAWING)

					PRA	VK/DV	YVR	WAM
0	14.08.2024	ISSUED FOR CONSTRUCTION (IFC)			PRA	VK/DV	YVR	WAM
B	27.05.2024	ISSUED FOR APPROVAL			PRA	KG/DV	YVR	WAM
A	13.01.2024	ISSUED FOR COMPANY REVIEW (ICR)			PT	KG/TS	YVR	WAM
REV	DATE	DESCRIPTION			DRAWN	CHKD	APPRD	APPROVED TARGET WAM
SCALE:	NTS	DRAWN:	PRA	CHKD:	VK/DV	APPRD:	YVR	APPRD TARGET WAM

أدنوك للغاز
ADNOC Gas



ADNOC

CONTRACTOR:

ENGINEERING SUBCONTRACTOR:	
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PROJECT TITLE:	HABSHAN & BAB CONTROL SYSTEMS UPGRADE PACKAGE 1
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DRAWING TITLE: PIPING AND INSTRUMENTATION DIAGRAM
FUEL AND STARTING GAS FOR
13-C-101A/B/102

UNIT: 13	PLANT: HABSHAN-1
CONTR. PROJ. No.: AET01074.00	CONT. DWG. No.: 3143235

ADNOC GAS AGREEMENT No.:		4700018747	
ADNOC GAS PROJ. No.:	ADNOC GAS DWG. No.:	SHEET No.:	REV

0405323	5323P1-H1-013-00-50-015	02/02	0
14	15	SHEET SIZE	

CONTRACT NO. 217128CD			BY	DATE	DR61 FUEL & STARTING GAS
TYPE DR513		DRAWN	SPN	07/21/92	DIAGRAM
SCALE NONE		CHECKED	JJM	07/21/92	DRESSER-RAND TURBO PRODUCTS DIVISION OLEAN, N.Y. USA
DEPT. 63.81		REDRAW			
DRAFTING APPROVAL		DJL	07/21/92		
CONTRACT APPROVAL		JAB	07/21/92		
SYSTEM APPROVAL		JJM	07/21/92		
LICENSEE					317-211
					SHT. REV. 202 04

CAD NO : 5222B141 012 00 50 015 02 DW/G